

Ryan Vuong

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EXPERIENCE

Software Development Engineer

July 2023 – Present

Amazon Web Services

Santa Clara, CA

- Owned and operated components of a large-scale distributed system vending authoritative networking data for all AWS data centers worldwide.
- Designed and implemented an SNS/SQS-based eventing pipeline for near real-time data propagation, reducing API call volume by 15%
- Built infrastructure to transform unstructured network entity data into optimized data models, enabling efficient API access and storage for 1B+ daily requests from internal AWS services.
- Extended enforced data schemas with historical audit metadata, enabling efficient storage and querying of entity change history for improved data integrity and analysis.
- Built an internal chatbot using AWS Strands and Bedrock, enabling secure, real-time querying of proprietary networking data via an internal MCP server.
- Authored and maintained thousands of lines of Infrastructure as Code (IaC) using AWS CDK and CloudFormation to provision and manage production-grade AWS resources, including IAM roles, Secrets Manager, S3, SQS/SNS, monitoring dashboards, Cloudwatch alarms, and deployment pipelines.
- Modernized the team's internal web portal by migrating from a REST-based frontend to a React and GraphQL architecture on a Flask backend, improving UI interactivity and enabling more flexible, efficient data fetching.
- Owned on-call responsibilities and conducted service audits to support global operations, resolving high-impact incidents and reducing recurring issues by 65%.
- Led a team-wide migration from long-term IAM user credentials to short-lived IAM role credentials (6-hour rotation), eliminating persistent secrets and reducing blast radius from credential exposure.

Software Development Engineer Intern

June 2022 – September 2022

Amazon Web Services

Seattle, WA

- Optimized full IPv4 scan of the internet, increasing the speed by 8x.
- Processed terabytes of IPv4 scan data using PySpark on AWS EMR to identify unresponsive address ranges.
- Automated large-scale data processing and storage using AWS Glue, S3, and Java, increasing IP probing response rates by 15%.
- Optimized AWS EC2 infrastructure by removing extraneous instances, resulting in 60% cost savings.

EDUCATION

University of California - Los Angeles

Los Angeles, CA

B.S. Computer Science

Sep. 2019 – June 2023

GPA: 3.76

Relevant Coursework: Operating Systems, Data Structures, Networks, Machine Learning,

TECHNICAL SKILLS

Languages: Python, Java, Ruby, SQL, JavaScript/TypeScript

Cloud & Infrastructure: AWS (DynamoDB, Lambda, S3, SNS, SQS, CloudWatch), CDK, Docker, IAM

Systems Experience: Distributed Systems, Data Modeling & Schema Design, Scalability & Performance Optimization

PROJECTS

Real-Time Trading Intelligence Platform | *Chrome Extension, Node.js, Python, SQLite*

- Built a Chrome extension that hooks into Robinhood's browser network layer to intercept and stream live market data in real time, capturing 50+ options contracts/sec with sub-2s latency and persisting locally in SQLite.
- Designed a Node.js HTTP server to ingest, deduplicate, and serve batched tick-level market data; developed a live dashboard with sortable tables, automatic refreshing, and real-time price highlighting.
- Engineered a streaming ML signal service in Python using LightGBM with Triple Barrier labeling that continuously processes real-time time-series price data with historical data (10GB+), computes multi-window momentum metrics and intraday volatility statistics, and surfaces actionable trading signals with confidence scores.